CS180 · PROJECT 3 · PART A

Fun with Flow Models

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From noise to images: denoising, guidance, editing, and visual ambiguity.

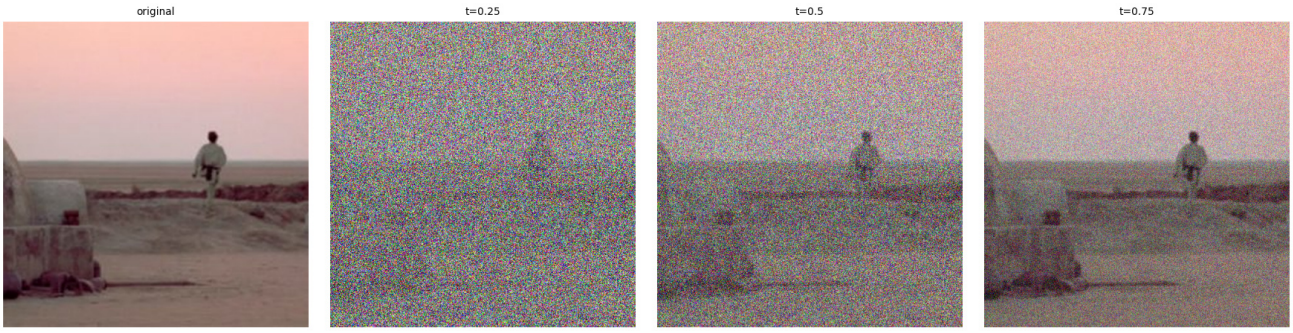
Portfolio content export. Figures and text match the website; pagination is adapted for readable submission.

PixNerd XXL · 512 × 512 · Euler step 0.02 · CFG weight 7

1.1 · From a photograph to noise

For sections 1.1 - 1.5 I used the Luke Skywalker desert sunset image. The input is center cropped to 512 x 512 and normalized to [-1, 1]. A single Gaussian noise tensor (seed 180) is reused all 3 times. This makes sure the differences only come from interpolation weight.

$$x_t = (1 - t)\varepsilon + t x_1$$



Original, then t = 0.25, 0.50, 0.75 (left to right).

At t = 0.25, noise obscures the high level details and the backgrounds. At t = 0.75, the overall horizon, silhouette and pinky sky are easy to recognize. Most fine details are also recognizable. As we increase time, we move more towards the clean image.

1.2 · A classical baseline

I first divide each noisy input by t in order to restore the image amplitude. Then I applied Gaussian blur for sigma values in (1, 2, 4, 8, 16). I chose the lowest MSE against the known original separately for each input. This uses the clean reference to select a baseline; and it is not blind denoising.

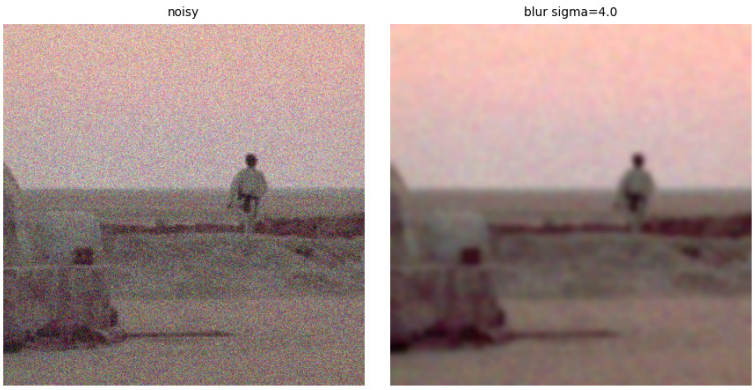
$\hat{x}\ 1 = G\ \sigma\ (x\ t\ /\ t)$



t = 0.25 · Noisy input (left), selected Gaussian blur (right). The selected σ appears above the result.



t = 0.50 · Noisy input (left), selected Gaussian blur (right). The selected σ appears above the result.



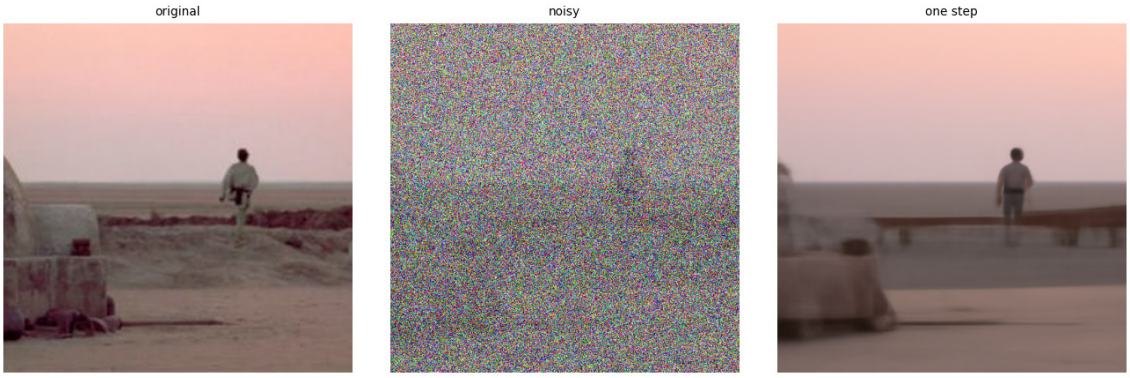
t = 0.75 · Noisy input (left), selected Gaussian blur (right). The selected σ appears above the result.

Averaging suppresses noise but also softens the person, buildings, and foreground texture. The flat sky is easier to recover than the small details.

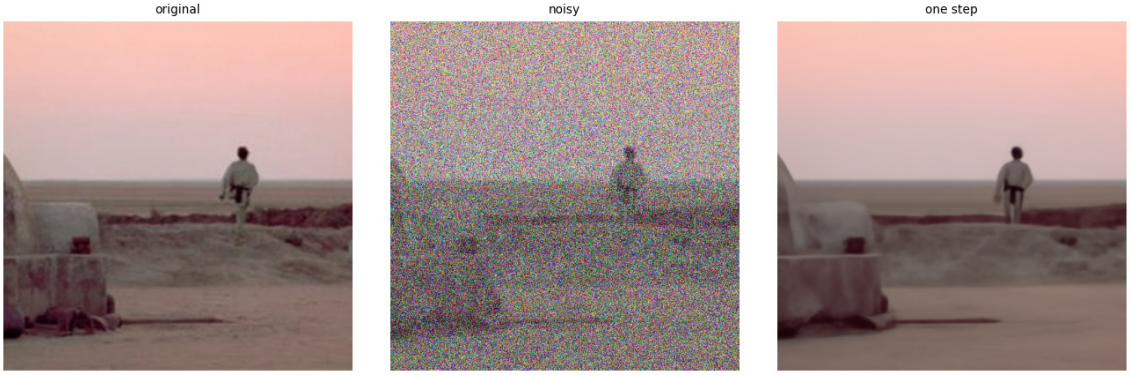
1.3 · One prediction toward the endpoint

We modify the pretrained model to predict velocity from the current image, the time, and a prompt - "a high quality photo". We estimate the clean endpoint in 1 shot after predicting the velocity.

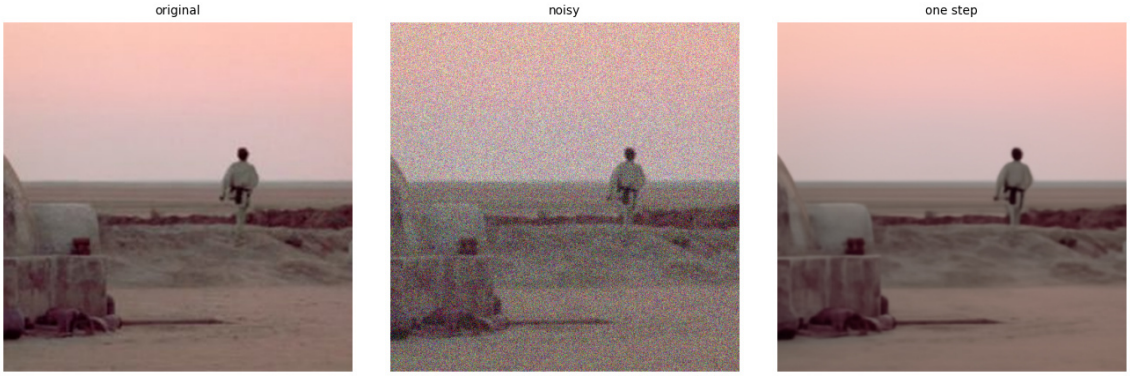
$\hat{x}\ 1 = x\ t + (1 - t)v\ \theta\ (x\ t\ ,\ t,\ c)$



t = 0.25 · Original, noisy input, one-step estimate (left to right).



t = 0.50 · Original, noisy input, one-step estimate (left to right).



t = 0.75 · Original, noisy input, one-step estimate (left to right).

The new learned estimate preserves more sharp scene components and structure compared to Gaussian blur. The larger the starting time, the more information about the image provided to the model -- the smaller the extrapolation the better it seems.

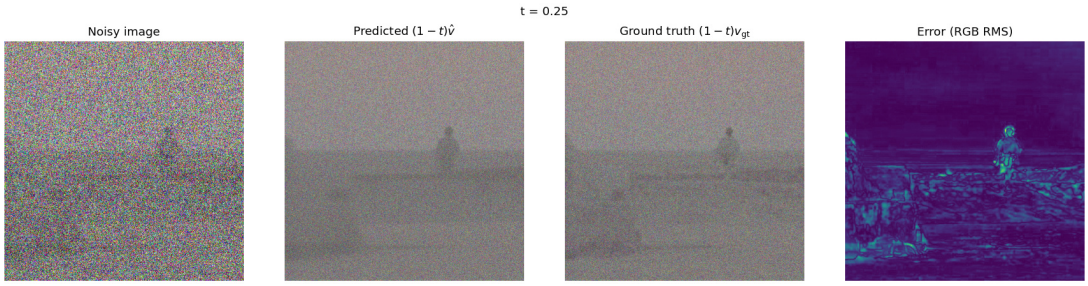
1.4 · What does the model add?

I compare the predicted updates with the exact updates using the same image and noise pair to keep things consistent. To measure the error, I calculate the squared difference at each pixel, average it across the RGB channels, and then take the square root. This is all done before any clipping or scaling for display.

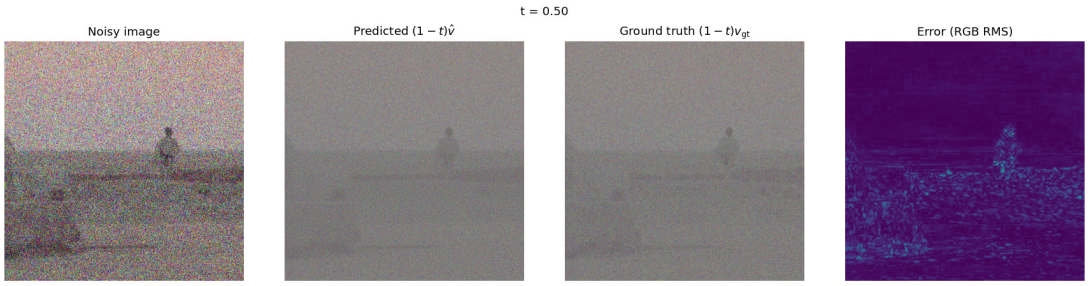
$\Delta = (1 - t)\hat{v}$ · $\Delta = (1 - t)(x\ 1 - \varepsilon)$ $E = \sqrt{\text{mean RGB}}\ [(\Delta - \Delta)^2]$



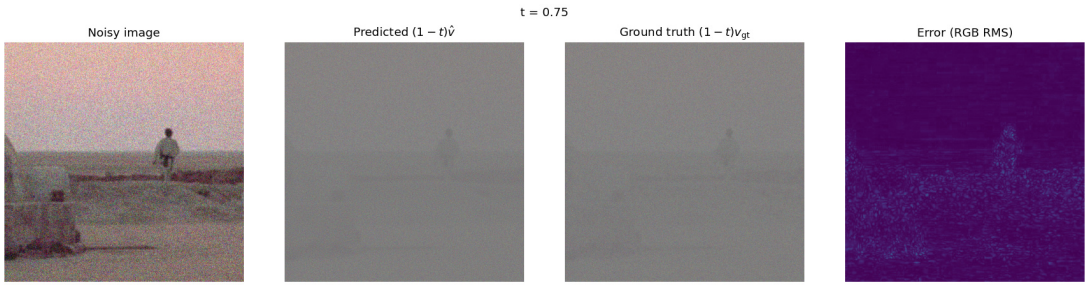
Original reference image.



t = 0.25 · Noisy input, predicted update, true update, RGB RMS error (left to right).



t = 0.50 · Noisy input, predicted update, true update, RGB RMS error (left to right).



t = 0.75 · Noisy input, predicted update, true update, RGB RMS error (left to right).

All updates use the same symmetric RGB display scale, with zero shown as middle gray. The error maps also share the same linear scale - it ranges from zero to the largest error across all three maps. Brighter areas == larger errors which are mostly concentrated around the figure and the textured foreground. The darker map at t = 0.75 shows that the update error is smaller at that point. Since the factor (1-t) also gets smaller, this doesn't necessarily mean the velocity pred is more accurate.

1.5 · Following the flow

Euler sampling predicts a new velocity at each state. We use step of h = 0.02.Since we use t = 0.50, we take 25 updates. We end once t >= 1.

$x_{t+h} = x_t + h v_\theta(x_t, t, c)$



Initial state and every fifth update: t = 0.50, 0.60, 0.70, 0.80, 0.90, 1.00 (row-major order).



Same t = 0.50 input · Original, Gaussian blur, one-step, Euler (left to right).

Reconstruction MSE in normalized tensor space; lower is better | Method | MSE | Gaussian blur | 0.00486194 | One-step | 0.00139233 | Euler | 0.00195181

Both one step and euler are clearly better than blur. One-step has the lowest MSE in this run, so Euler is not automatically better. Integrating a learned field means following its preferred image trajectory, and this can lead to images that are not exactly similar to the original. However, I do believe if our image had much more finer details, the Euler reconstruction would be most accurate.

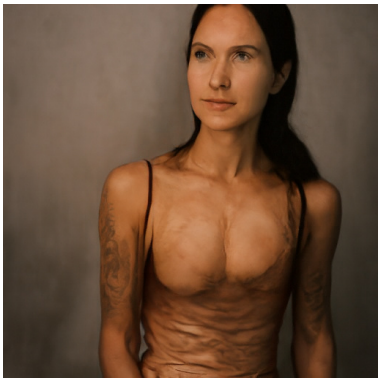
1.6 · Images from pure noise

I've performed 50 euler updates for our five gaussian tensors using "a high quality photo" prompt. We start at t = 0 and use Seeds 180-184 for the CFG comparison.

Our prompt is generic, but all five samples are still portraits. Seed 180 has a featureless face and Seed 183 has rough painterly facial artifacts. Other samples are much more coherent, but it is clear that this small set shows strong portrait preference.



Seed 180 · Ordinary conditional sampling



Seed 181 · Ordinary conditional sampling



Seed 182 · Ordinary conditional sampling



Seed 183 · Ordinary conditional sampling



Seed 184 · Ordinary conditional sampling

2.1 · Strengthening the condition

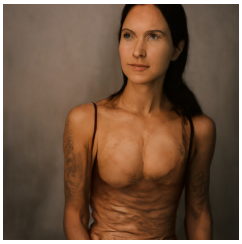
We use CFG to combine predictions from the prompt - "a high quality photo" and an empty prompt. For Guidance we use w = 7 and h = 0.02.

v CFG = v c + 7(v c − v u)

Due to the extra strength from CFG, we get much more clear facial features and framing. I think this is best shown by seeds 180 and 183. The main take away is that even though the samples are still different in appearance, their general composition is very similar.



Seed 180 · Without CFG



Seed 181 · Without CFG



Seed 182 · Without CFG



Seed 183 · Without CFG



Seed 184 · Without CFG



Seed 180 · CFG, w = 7



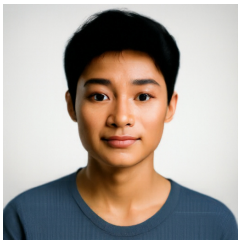
Seed 181 · CFG, w = 7



Seed 182 · CFG, w = 7



Seed 183 · CFG, w = 7



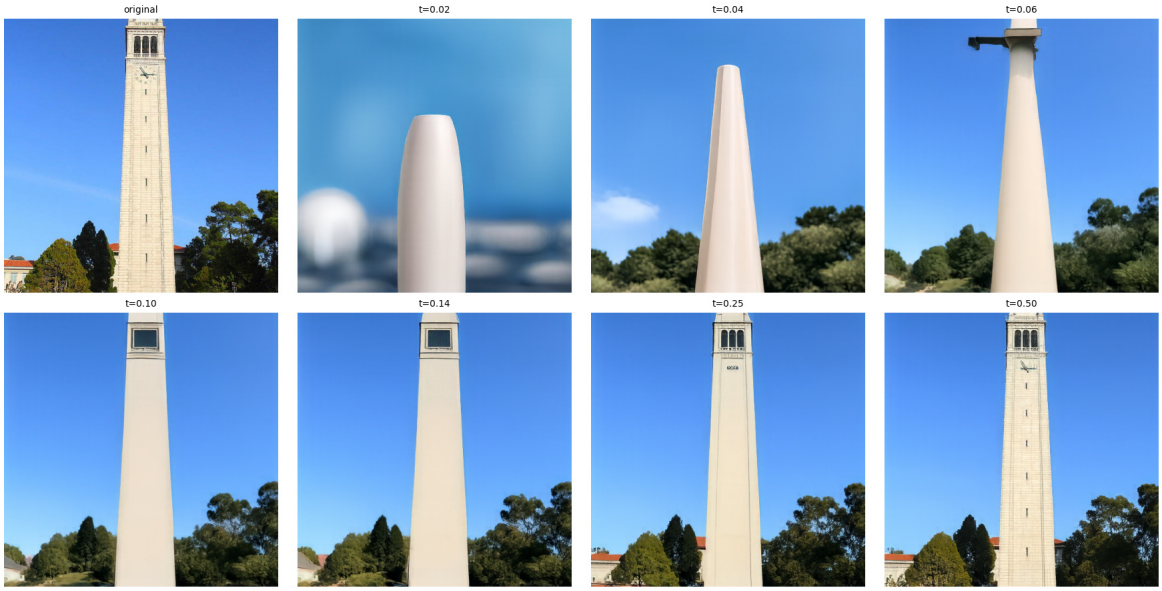
Seed 184 · CFG, w = 7

3.1 · How much of an image survives?

I mix each input with fixed seed-180 noise, and then use CFG - “a high quality photo”. Every sweep used h = 0.02 and w = 7. Lower starting times remove more source information and allow stronger reinterpretation.

t start ∈ {0.02, 0.04, 0.06, 0.10, 0.14, 0.25, 0.50}

Campanile

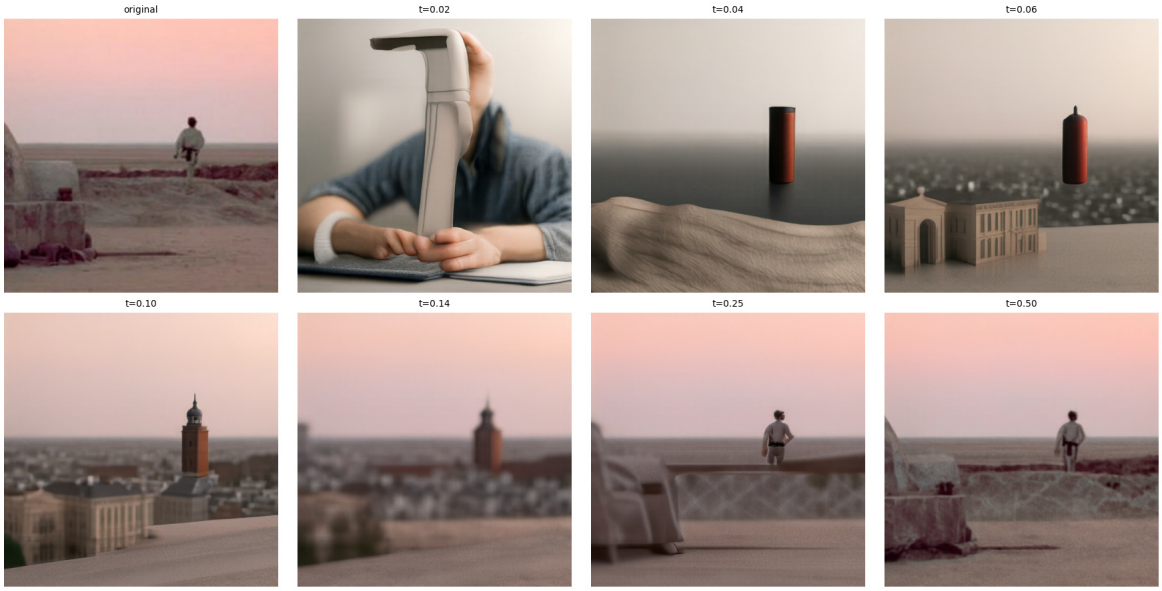


Original, then t = 0.02, 0.04, 0.06, 0.10, 0.14, 0.25, 0.50, read left to right across each row.

At low starting times, the clock tower gets reinterpreted to a 3d tower, and then to a lighthouse. The details then end up returning.

3.1 · How much of an image survives?

Luke at sunset

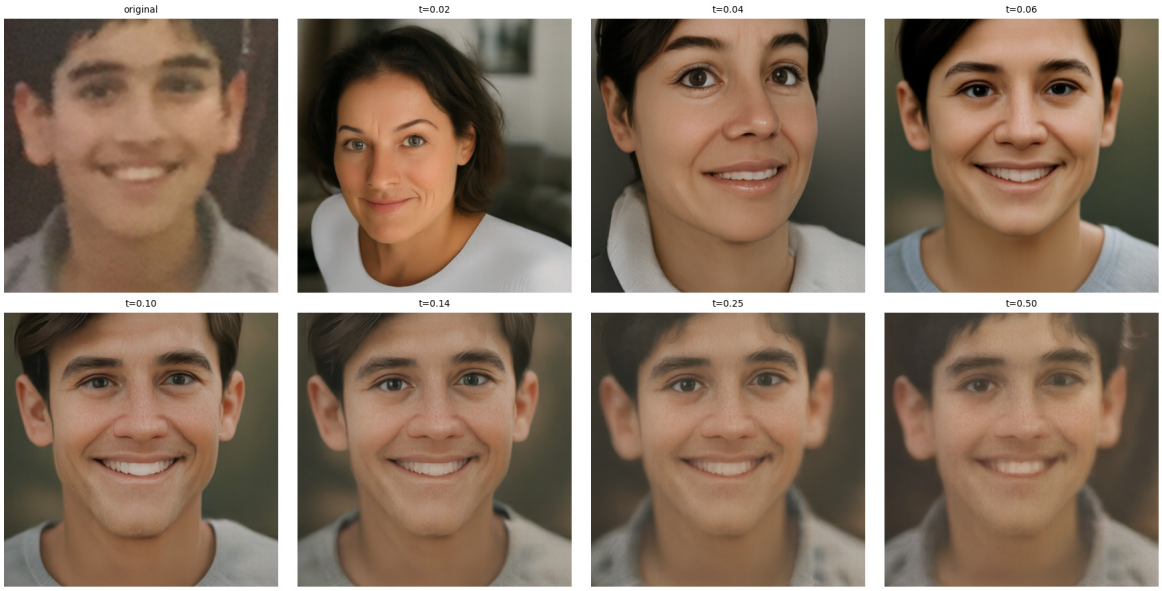


Original, then t = 0.02, 0.04, 0.06, 0.10, 0.14, 0.25, 0.50, read left to right across each row.

At t = 0.02, the model changes the scene into a close-up that looks nothing like the original. At intermediate values, the standing figure starts looking more like a tall structure. By t = 0.25-0.50, the original scene becomes recognizable again.

3.1 · How much of an image survives?

Childhood portrait



Original, then t = 0.02, 0.04, 0.06, 0.10, 0.14, 0.25, 0.50, read left to right across each row.

At lower starting times, the model changes the person's face quite a bit. As t increases, the edits become less noticeable. The original smile and framing stay mostly intact at higher values.

3.1 · How much of an image survives?

Web illustration · Yoda

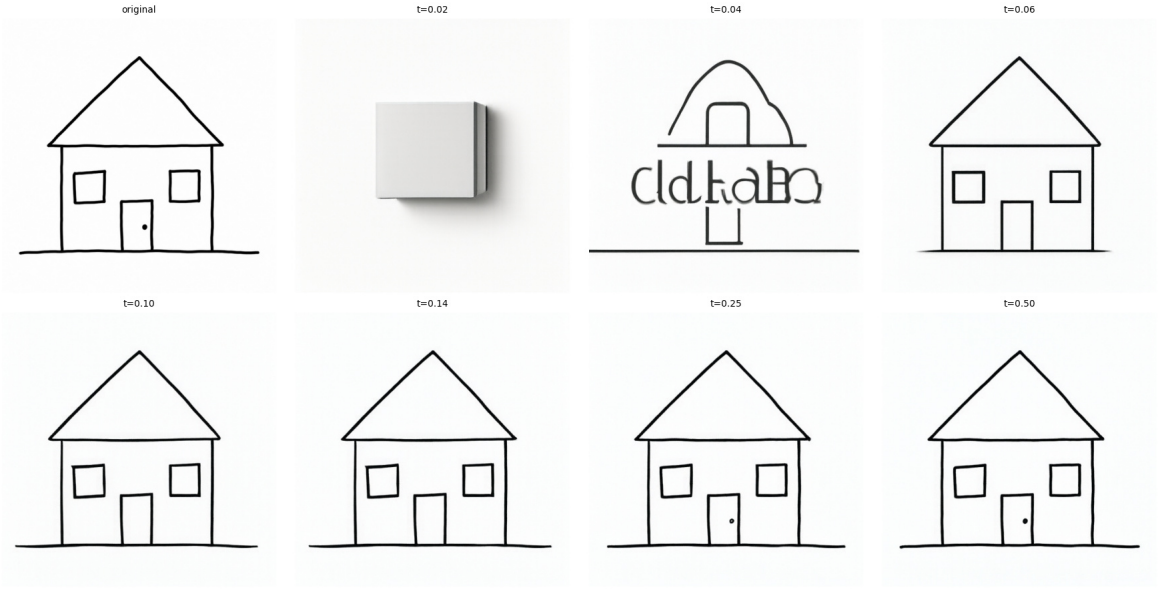


Original, then t = 0.02, 0.04, 0.06, 0.10, 0.14, 0.25, 0.50, read left to right across each row.

The strongest edit turns the illustration into a portrait. At intermediate times, the model keeps the red background and dark silhouette but changes how the scene looks. With higher values of t, the result stays closer to the original illustration instead of becoming a realistic photo.

3.1 · How much of an image survives?

Hand drawing · House

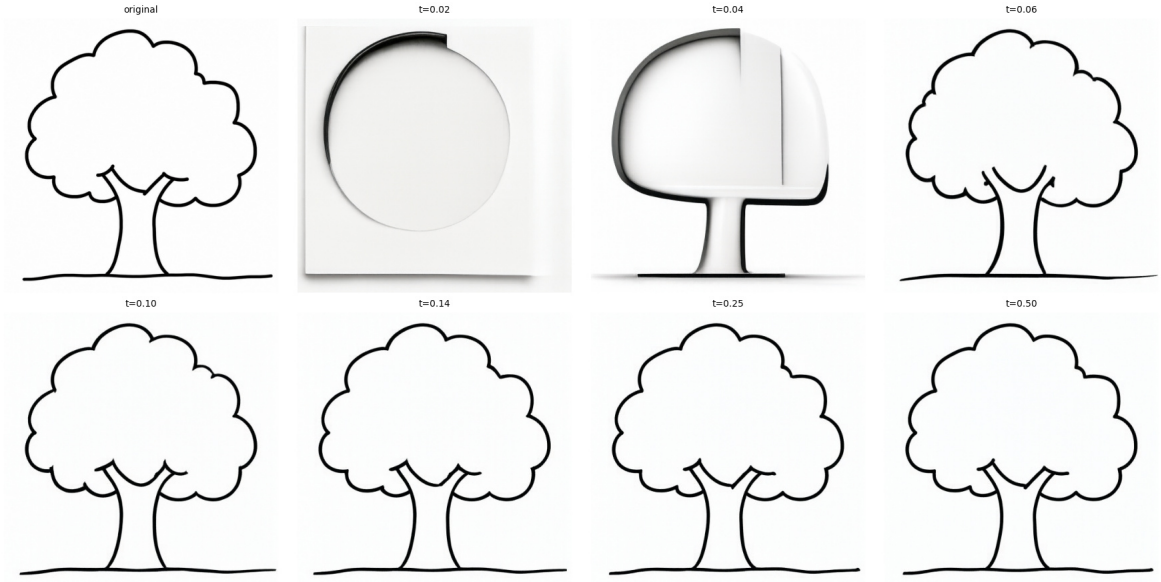


Original, then t = 0.02, 0.04, 0.06, 0.10, 0.14, 0.25, 0.50, read left to right across each row.

At lower values of t, the model produces unrelated objects or distorts the original lines. Once t reaches around 0.06, the house becomes recognizable again. Even with a photo prompt, the model struggles to turn the simple sketch into a realistic image.

3.1 · How much of an image survives?

Hand drawing · Tree



Original, then t = 0.02, 0.04, 0.06, 0.10, 0.14, 0.25, 0.50, read left to right across each row.

At t = 0.02-0.04, the model turns the tree outline into smooth, unrelated shapes. By t = 0.06, the tree starts looking recognizable again. At higher values, the results stay pretty close to my original drawing.

3.2 · Editing with a destination in mind

For this part, I used the same three images, noise seed, and seven starting times. I only changed the text prompt to see how much the model would follow the new description while keeping the original image.

Campanile → rocket

a photograph of a tall rocket on a launch pad surrounded by trees under a blue sky



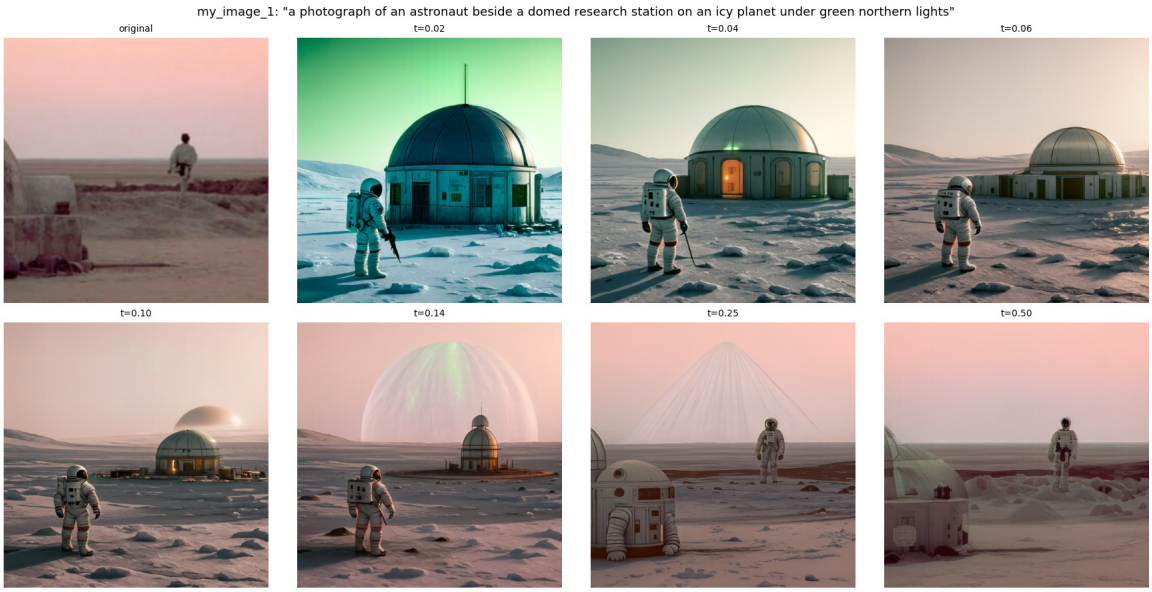
Original, then t = 0.02, 0.04, 0.06, 0.10, 0.14, 0.25, 0.50, read left to right across each row.

At lower starting times, the Campanile turns into a rocket pretty clearly. Around t = 0.10-0.14, the results mix the tower's shape with rocket-like details. By t = 0.50, the original Campanile becomes the main focus again.

3.2 · Editing with a destination in mind

Desert → icy planet

a photograph of an astronaut beside a domed research station on an icy planet under green northern lights



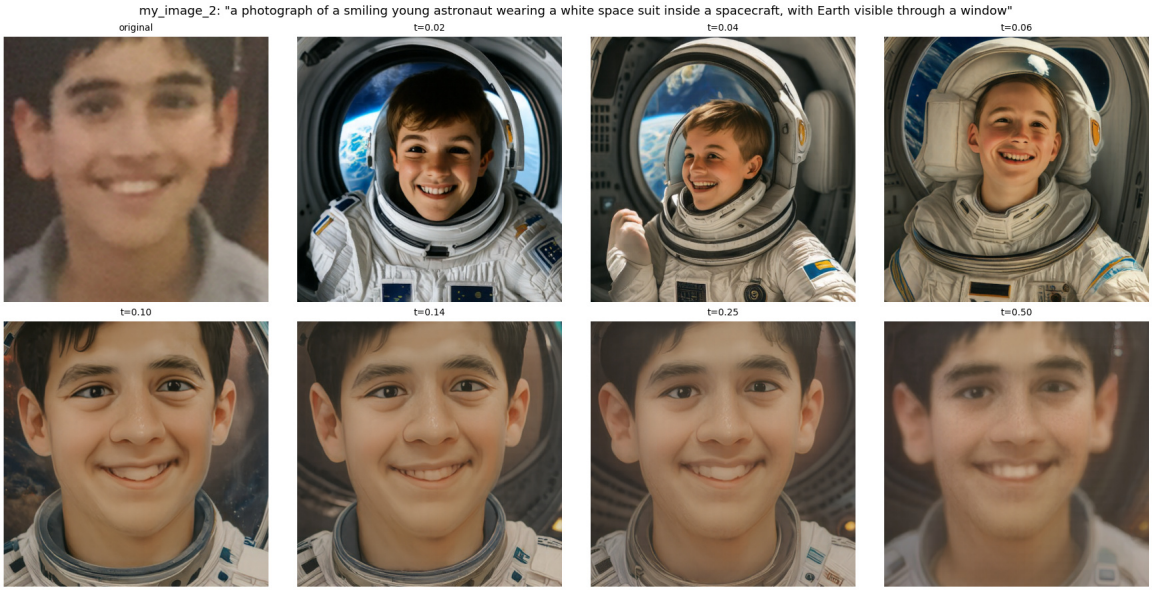
Original, then t = 0.02, 0.04, 0.06, 0.10, 0.14, 0.25, 0.50, read left to right across each row.

The earlier results show an astronaut and a dome-shaped station, with the strongest green lighting at t = 0.02. As t increases, the model keeps more of the original desert sunset and layout. Because of this, the later images don't fully match the icy-planet prompt.

3.2 · Editing with a destination in mind

Portrait → astronaut

a photograph of a smiling young astronaut wearing a white space suit inside a spacecraft, with Earth visible through a window



Original, then t = 0.02, 0.04, 0.06, 0.10, 0.14, 0.25, 0.50, read left to right across each row.

The first three edits turn the person into an astronaut wearing a helmet, although the face looks quite different. At higher starting times, the original face is easier to recognize, but the astronaut outfit becomes less noticeable. The model also struggles to consistently include Earth through the window.

3.3 · Two readings, one image

For each Euler step, I calculate the guided velocity for the upright image and then do the same for the image rotated by 180°. I rotate the second velocity back and average the two predictions. The rotation only affects the image's height and width, leaving the batch and color channels unchanged.

$v = \frac{1}{2}[v_{CFG}(x,t,c_a) + F(v_{CFG}(F(x),t,c_b))]$

Wizard / castle



Upright · an oil painting of a bearded wizard's face



Rotated 180° · an oil painting of a mountain castle above a river

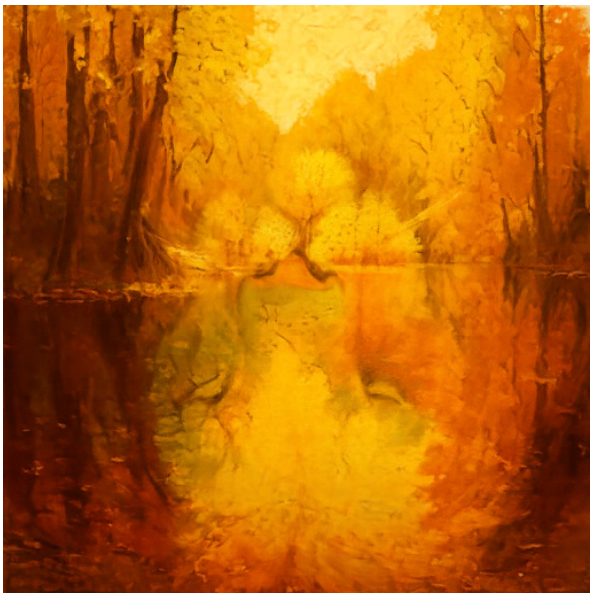
The wizard is easy to recognize when the image is upright. After rotating it, the beard and hat start resembling parts of a landscape. However, the castle isn't very clear, so the illusion only works partially.

3.3 · Two readings, one image

Lion / autumn forest



Upright · an oil painting of a lion's face



Rotated 180° · an oil painting of a golden autumn forest reflected in a lake

The upright image clearly shows a lion. When rotated, the darker vertical shapes start looking like trees surrounding a bright lake reflection. This creates a more convincing illusion than the wizard and castle example.

3.4 · Different subjects at different scales

I combine the low-frequency parts of the distant prompt's guided velocity with the high-frequency parts of the close-up prompt's velocity. Both predictions use the same image and time step. For the Gaussian filter, I use a kernel size of 265 and a sigma of 16 on a 512 × 512 image.

$$v = G_{16}(v_a) + v_b - G_{16}(v_b)$$

Lion / sunflowers



Close · a lithograph of a field of sunflowers



Distant (64 × 64) · a lithograph of a lion's face

Up close, the sunflower heads and stems are easy to see. From farther away, the larger shapes start forming something that resembles a face, although the lion isn't very clear. This illusion works less effectively than the other example and could probably improve with a different prompt or random seed.

3.4 · Different subjects at different scales

Sea turtle / ancient city



Close · a lithograph of an ancient city with winding cobblestone streets



Distant (64 × 64) · a lithograph of a sea turtle seen from above

Up close, the image shows buildings and winding paths that resemble an ancient city. From a distance, the overall shape starts looking like a sea turtle, with a shell and flippers. The city details are still visible, though, so the illusion doesn't completely switch between the two subjects.